

ICT 1201

Fundamentals of Computer Systems

Introduction to Operating Systems

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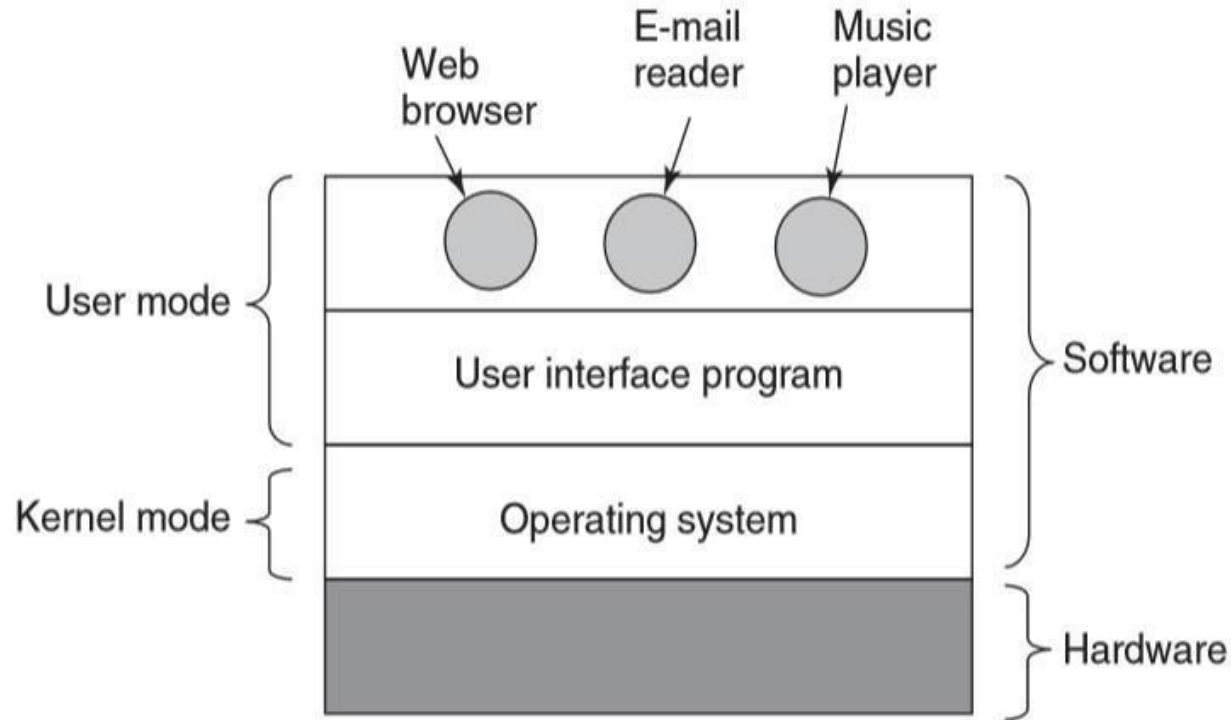
What is an Operating System?

- ❑ A modern computer consist of
 - ❑ One or more processors
 - ❑ Main memory
 - ❑ Disks
 - ❑ Printers
 - ❑ Various input/output devices
- ❑ Writing programs that keep track of all these components and use them correctly, let alone optimally, is an difficult job.

What is an Operating System?

- ❑ For this reason, computers are equipped with a layer of software called operating system.
- ❑ An **operating system** is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of program.

What is an Operating System?



What is an Operating System?

- Examples
 - Windows
 - Linux
 - Mac OS



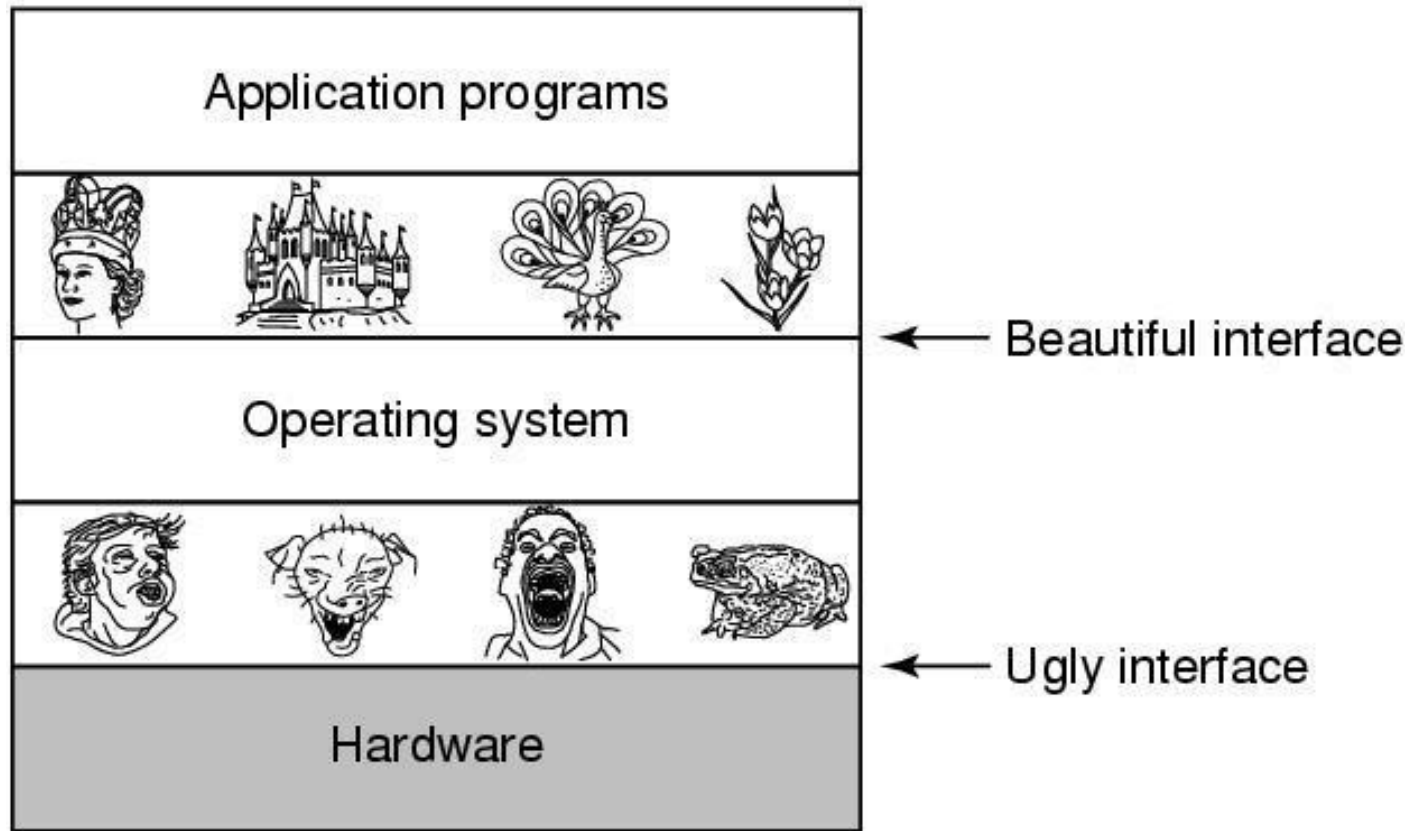
Operating System as an Extended Machine

- ❑ The architecture (instruction set, memory organization, I/O and bus structure) of most computers at the machine language level is primitive and awkward to program.
- ❑ Users does not want to involved with the programming hardware.
- ❑ Instead, what the user want is a simple, high level abstraction to deal with.
 - In the case of disks, a typical abstraction would be that the disk contains a collection of named files, each file can be open, read, write and close.

Operating System as an Extended Machine

- ❑ The program that hides the truth about the hardware from the user and present a nice, simple view of named files that can be read and written is the Operating System.
- ❑ The abstraction offered by the operating system is simpler and easier to use than that offered by the underlying hardware.

Operating System as an Extended Machine



Operating System as a Resource Manager

- ❑ A modern computer consists of one or more processors, memories, disks, printers and wide variety of other devices.
- ❑ The job of the operating system is to provide for an orderly and controlled allocation of the above devices among the various programs competing for them.

Operating System as a Resource Manager

- ❑ Operating system need to keep track of who is using which resources
 - To grant recourse request,
 - To account for usage,
 - And to mediate conflicting requests.
- ❑ Resource sharing (multiplexing) can be done in two ways.
 - In time
 - In space

Operating System as a Resource Manager

- In **time multiplexing**, different programs take turns to use resources. First one gets to use the resource, then another, and so on.
- Operating system determine who goes next and for how long.

Operating System as a Resource Manager

- ❑ In **space multiplexing**, each program gets part of the resources.
- ❑ For example, main memory divided up among several running programs and it is more efficient to hold several programs in memory at once rather than giving it whole to one program.

Functions of an Operating System

- ❑ Memory Management
- ❑ Processor Management
- ❑ Device Management
- ❑ File Management
- ❑ Security
- ❑ Control over system performance
- ❑ Job accounting
- ❑ Error detecting aids
- ❑ Coordination between other software and users.

Memory Management

- ❑ Main memory provides a fast storage that can be accessed directly by the CPU. For a program to be executed, it must be in the main memory.
- ❑ An Operating System does the following activities for memory management.
 - Keep track of primary memory : what part of it are in use by whom, what part is not.
 - In multiprogramming, the OS decides which process will get memory when and how much.
 - Allocate and de-allocates the memory.

Processor Management

- ❑ In multiprogramming environment, the OS decides which process gets the processor when and for how long. This function is called **process scheduling**.
- ❑ An Operating System does the following activities for processor management.
 - Keeps track of processor and its status. The program responsible for this task is known as **traffic controller**.
 - Allocates and de-allocates the CPU(processor) to a process

Device Management

- An Operating System manages device communication via their respective drivers. It does the following activities for device management.
 - Keep track of all devices. The program responsible for this task is known as the **I/O controller**.
 - Decides which process gets the device when and for how long.
 - Allocates and de-allocates the device in most efficient manner.

File Management

- ❑ A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directories.
- ❑ An Operating System does the following activities for file management.
- ❑ Keeps track of information, location, uses, status etc. The collective facilities are often known as **file system**.
- ❑ Decides who gets the resources.
- ❑ Allocates and de-allocates the resources

Other Important Activities

- ❑ Security
 - ❑ By means of password and similar other techniques, it prevent unauthorized access to programs and data.
- ❑ Control over system performance
 - ❑ Recording delays between request for a service and response from the system.
- ❑ Job accounting
 - ❑ Keeping track of time and resources used by various jobs and users.

Other Important Activities

- ❑ Error detecting aids
 - ❑ Production of trace, error message and other debugging and error detecting aids.
- ❑ Coordination between other software and users
 - ❑ Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer system.

Shell

- ❑ In computing, a shell is a user interface for access to an operating system's services.
- ❑ Two types
 - ❑ Command line interface(CLI)
 - ❑ Graphical user interface(GUI)

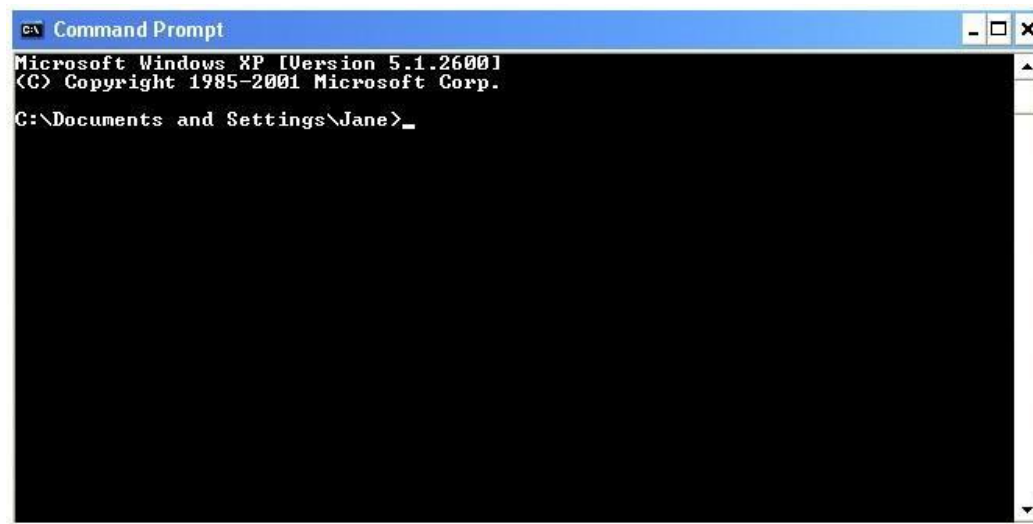
Graphical User Interface (GUI)

- ❑ Is a type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators.
- ❑ For example, all versions of Microsoft Windows utilize a GUI.



Command Line Interface (CLI)

- ❑ Is a text-based interface that is used to operate software and operating system while allowing user to respond to visual prompts by typing single commands into the interface and receiving a reply in the same way.
 - E.g. DOS, Linux, UNIX



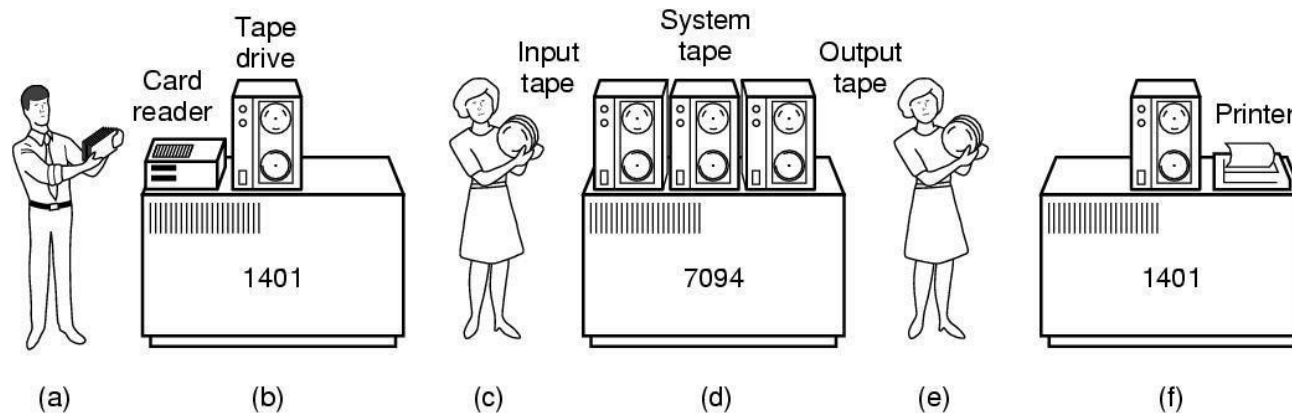
Types of Operating Systems

❑ Batch Operating System

- ❑ The users of batch processing system do not interact with the computer directly. Each user prepares his job on punch cards and submit it to the computer operator.
- ❑ To speed up processing, jobs with similar needs are batched together and run as a group.

Types of Operating Systems

- Batch Operating System



- (a) Programmers bring cards to 1401.
- (b) 1401 reads batch of jobs on to tape.
- (c) Operator carries input tape to 7094.
- (d) 7094 does computing.
- (e) Operator carries output tape to 1401.
- (f) 1401 prints output.

Types of Operating Systems

❑ Batch Operating System

❑ An operating system does the following activities related to batch processing.

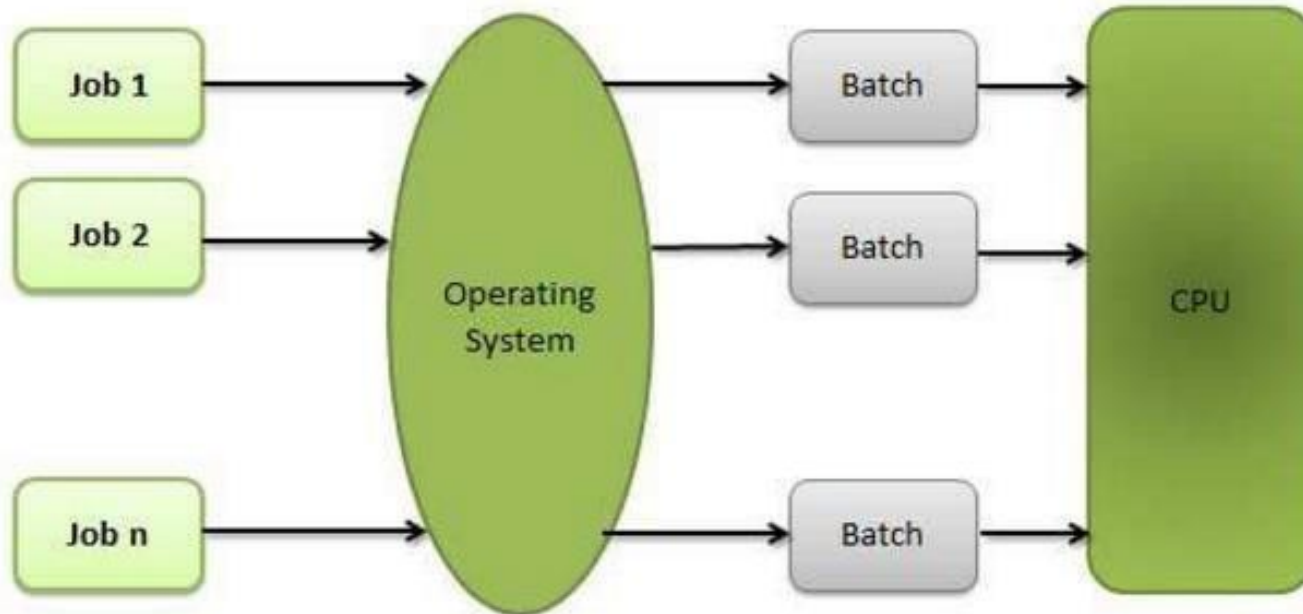
- OS collects one programs and data together in a batch before processing starts.
- Jobs are processed in the order of the submission.
- When job completes its execution, its memory is release and the output for the jobs gets copied into an output spool for later printing or processing.

Types of Operating Systems

- ❑ Batch Operating System
 - ❑ Spooling refers to putting data of various I/O jobs in a buffer.
 - ❑ This buffer is a special area in memory or hard disk which is accessible to I/O devices.

Types of Operating Systems

❑ Batch Operating System



Types of Operating Systems

- ❑ Time-sharing Operating System
 - ❑ Time-sharing is a technique which enables many people, located at various terminals, to use a particular computer system at the same time.
 - ❑ Processor's time which is shared among multiple users simultaneously is termed as time-sharing.
 - ❑ The OS uses CPU scheduling and multiprogramming.
 - ❑ Objective is to minimize response time.

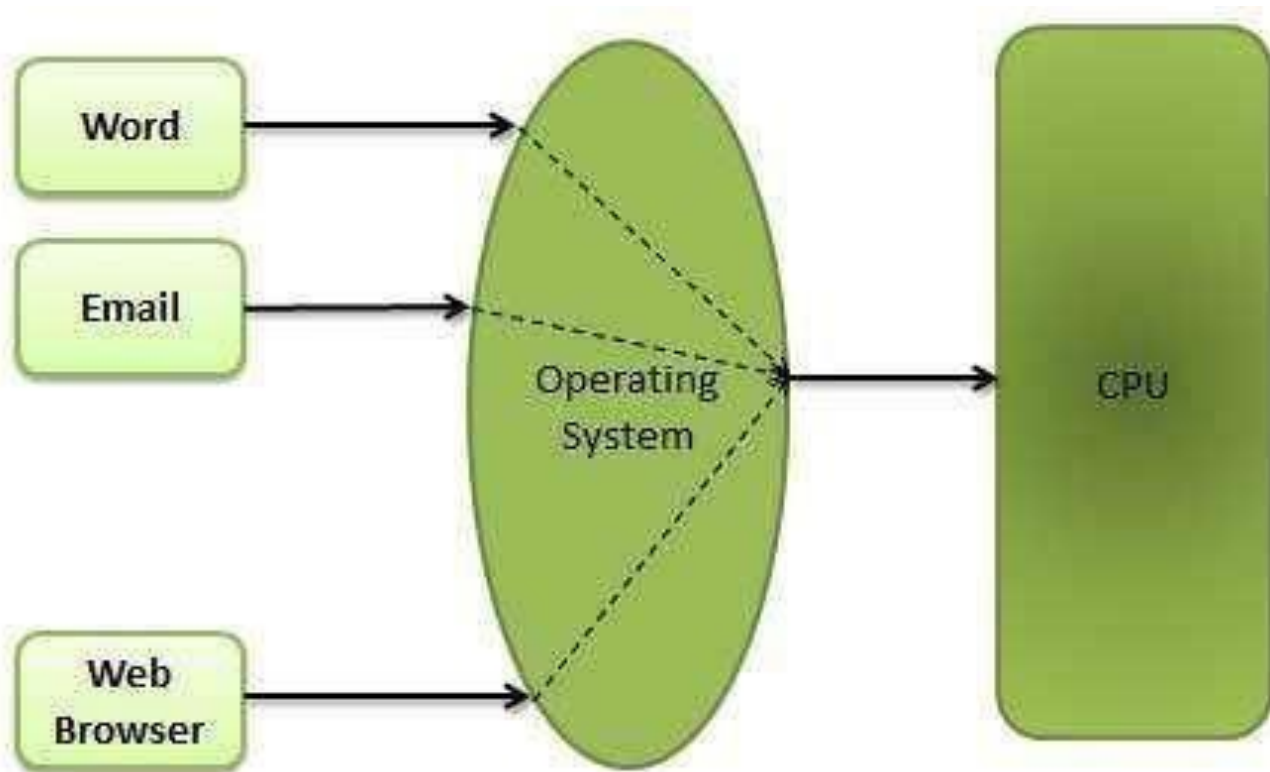
Types of Operating Systems

❑ Time-sharing Operating System

- ❑ An OS does the following activities related to Time-sharing / multitasking.
 - A program that is loaded into memory and is executing is commonly referred to as a process.
 - When a process executes, it typically executes for only a very short time before it either finishes or needs to perform I/O.
 - Since interactive I/O typically runs at slower speeds, it may take long time to complete.
 - During this time, a CPU can be utilized by another process.

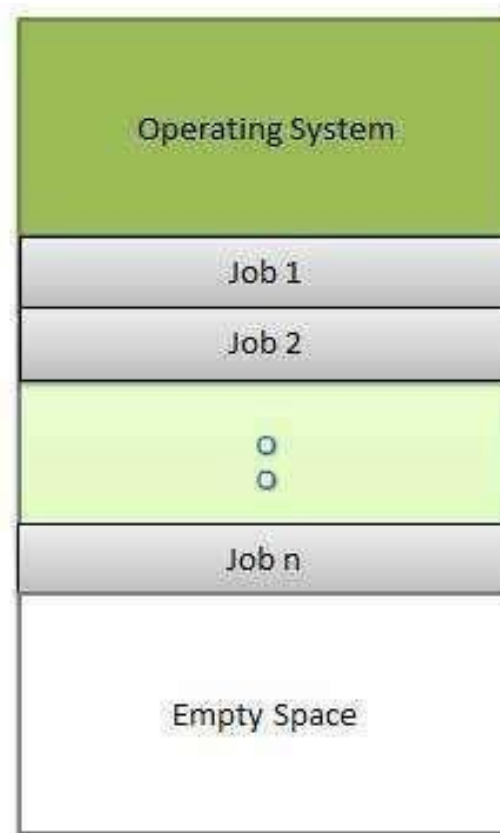
Types of Operating Systems

❑ Time-sharing Operating System



Types of Operating Systems

- ❑ Memory layout for multi-tasking/time-sharing/multi-programming

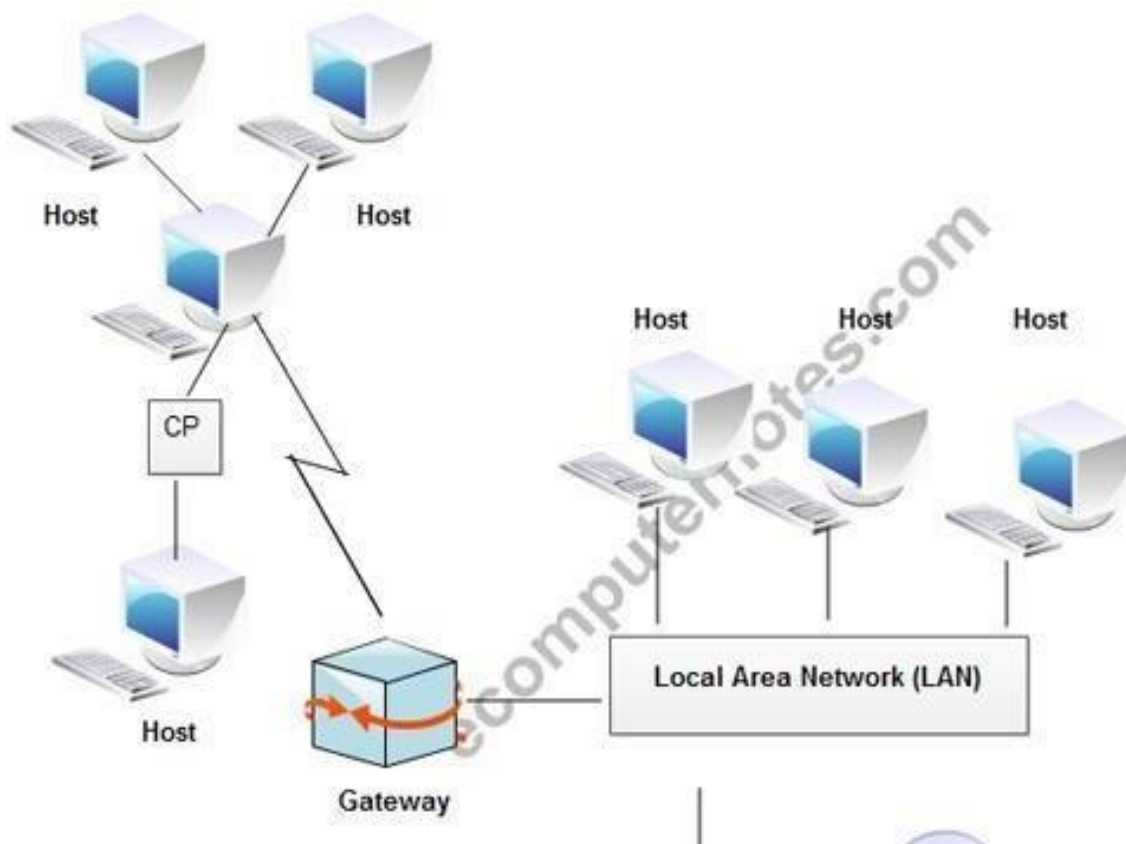


Types of Operating Systems

- ❑ Distributed Operating System
 - ❑ Distributed systems use multiple central processors to serve multiple real-time applications and multiple users.
 - ❑ Collection of independent, networked, communicating and physically separate Computational nodes.
 - ❑ Data processing jobs are distributed among the processors.
 - ❑ The processors communicate with one another through various communication lines. E.g. high speed buses, telephone lines
 - ❑ These also referred as loosely coupled systems.

Types of Operating Systems

❑ Distributed Operating System



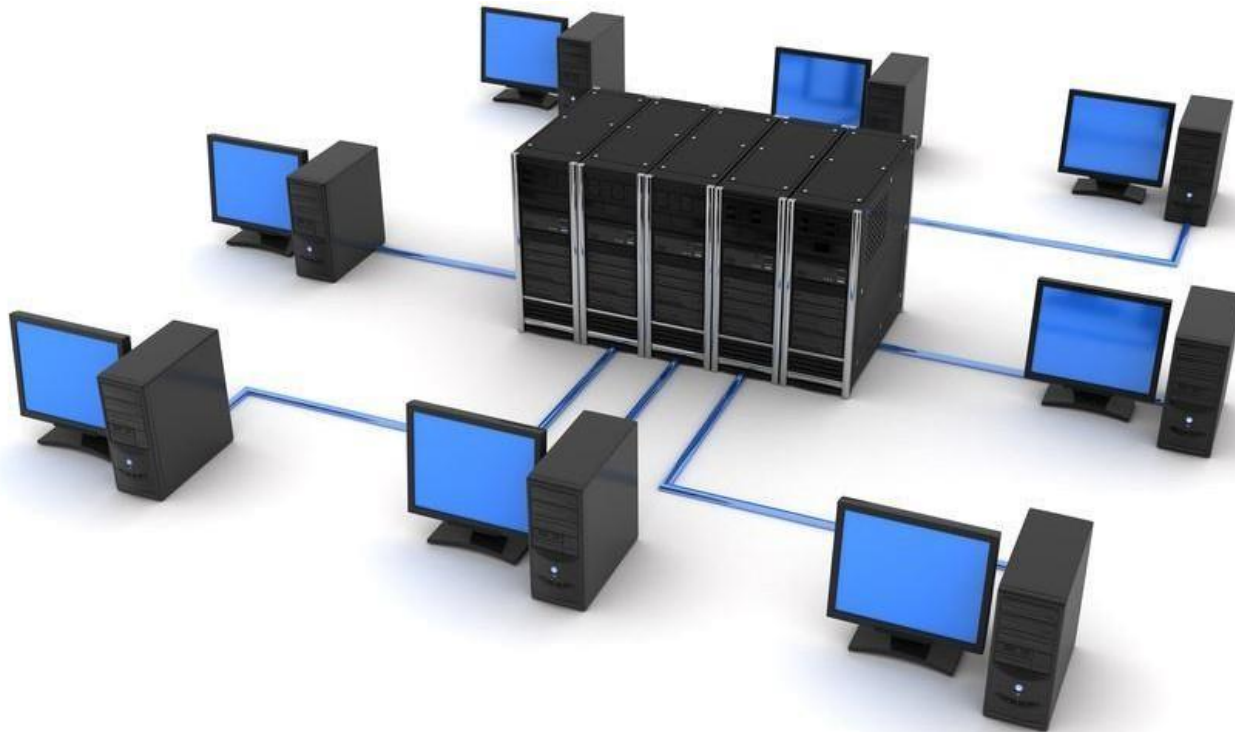
Types of Operating Systems

❑ Network Operating System

- ❑ A network Operating System runs on a server and provides the server the capability to manage data, users, security and application.
- ❑ The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network.
- ❑ E.g. Microsoft Windows server 2003, 2008, UNIXX, Linux, Mac OSX, BSD.

Types of Operating Systems

❑ Network Operating System



Types of Operating Systems

❑ Real-Time Operating System

- ❑ This is defined as a data processing system in which time interval required to process and respond to inputs is so small that it controls the environment.
- ❑ The time taken by the system to respond to an input and display of required updated information is termed as the response time.
- ❑ A real-time operating systems must have well defined, fixed time constraints, otherwise the system will fail.
 - ❑ E.g. Scientific experiments, weapon system, air traffic control systems

Types of Operating Systems

- ❑ Realtime Operating System
 - ❑ There are two types of realtime operating systems.
 - ❑ Hard real time systems
 - ❑ Guarantee that critical tasks complete on time.
 - ❑ Soft real time systems
 - ❑ Less restrictive.
 - ❑ A critical real time task get priority over other tasks and retains the priority until it complete

Types of Operating Systems

❑ Mobile Operating System

❑ Specifically designed to run on mobile devices such as mobile phones, PDAs, tablet computers and other handheld devices.

- Android OS
- BlackBerry OS
- Symbian OS
- iOS
- webOS
- MeeGO OS

MS – DOS OS

- ❑ MS-DOS (Microsoft Disk Operation System) is a single – user, single – tasking computer operating system that uses a non – graphical command line interface.
- ❑ DOS uses a directory system, where all the files are contained within a particular directory or a sub directory.



Windows OS

- ❑ Is a graphical operating system developed, marketed and sold by Microsoft.
- ❑ Is a commercial product.
- ❑ Eliminated the need for remembering the codes and the actions that go with the code.



DOS vs Windows

Feature	DOS	Windows
GUI	Used a text based interface	Uses graphics, images and text
Input System	Text is used as the basic input system commands	Uses a mouse for all OS input
Multitasking	Unable to run multiple processes at the same time	Allow more than one process to work simultaneously
Storage Size	The highest amount of storage size available is 2GB	Offer storage space up to 2terabyte
Price	DOS is free	Windows is costly

Linux



- ❑ Linux is an open source, crossplatform operating system.
- ❑ Some of the popular Linux distributions (Linux based OS) include Ubuntu, Fedora and Debian.
- ❑ Users can even alter the kernel in order to make changes to the operating system itself.
- ❑ Having full access to the code helps bring about faster improvements and bug fixes.

Homework

- “Linux is better than Windows”. Do you agree with this statement? If so provide necessary points to prove it.

THANK YOU